

Mykola Zhuk

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📁 Experience

Software Developer - Client MVP

Jun 2026 – Present | Remote

Independent project for a startup founder

- **Independently built** a Python pipeline covering 20 corporate-event categories. Processed 2,936 articles into 1,332 records, with **99.7% passing schema validation**.
- **Deployed and demonstrated the MVP on Azure** using Container Apps Jobs, Service Bus, Blob Storage, and PostgreSQL-backed idempotency. Added tests, audit logs, and inference-budget limits.
- Selected Mistral Small 3.2 24B through project-specific model evaluation; **reduced token usage by 17.9%** through cached classification, candidate filtering, and selective model calls.
- Improved candidate **precision from 45.0% to 58.6%** at approximately 95% recall on a 300-article holdout.

Backend Engineer - Volunteer

Oct 2025 – Mar 2026 | Remote

Tacmedbat

- Built NestJS/MongoDB APIs and a custom admin interface enabling the customer to edit Ukrainian/English landing-page content. Designed a public endpoint combining **8 collections to support 10+ frontend blocks**.
- Delivered **15+ features and fixes** across image handling, access control, payments, and SMTP delivery for contact forms.

Software Analyst

Jun 2024 – Aug 2024 | Toronto

Accenture

- Developed Python/OpenAI API prototypes for retail Q&A and semantic search, supported by data pipelines ingesting and enriching **50,000+ product records**.
- Delivered features with a cross-functional team through Agile sprints, code reviews, and stakeholder demos.

📁 Projects

Full-Stack Engineer

Oct 2024 – Present | Toronto

Scriptorium-CSC309 Project

- Built template, viewing, forking, and code-execution **APIs and frontend flows** using Next.js, React, Prisma, and PostgreSQL within a project team.
- Built Docker language environments with **execution-time and memory limits for 10+ languages**. Deployed the application on AWS EC2.
- Added **GitHub and Google OAuth 2.0 sign-in**. Tested APIs with Postman and checked responsive layouts.
- **Live Demo (AWS)**

Backend Engineer

Oct 2024 – Dec 2024 | Toronto

ProjectOllon (Industry-Partnered MVP) - CSC301 Project

- Designed a Prisma schema for **100+ files and 2,000+ code components, reducing full-scan time by 80%** in project evaluation.
- Integrated static analysis with OpenAI/Cohere models to flag duplication and complexity; helped deliver the code-review MVP to industry partner Ollon.

Machine Learning Engineer

Mar 2025 – Apr 2025 | Toronto

ML Classifier- CSC311 Project (scikit-learn, Pandas, PyTorch)

- Compared Random Forest and PyTorch models on **1,644 survey rows**, with preprocessing and respondent-grouped splits.
- Achieved **89.96% accuracy on 249 held-out rows**. Packaged the model and preprocessing for CLI prediction.

🧠 Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, Bash

Frameworks & Data: React, Next.js, Node.js, NestJS, PostgreSQL, MongoDB, Prisma, PyTorch, scikit-learn

Cloud & testing: Azure, AWS EC2, Docker, GitHub Actions, pytest, Jest, Grafana Loki, Postman, Git, Linux

🎓 Education

Honours B.Sc, Mathematics

Jun 2025

University of Toronto